SUPERMART

A Cross-Platform Grocery & Retail Mobile/Web Application

Unified Customer, Admin & Staff Multitenant Architecture

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MINI LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course **CSE315: Software Engineering in the Computer Science and Engineering Department**

DAFFODIL INTERNATIONAL UNIVERSITY

Dhaka, Bangladesh

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DECLARATION

We hereby declare that this lab project has been done by us under the supervision of **Dr. Mohammad Nuruzzaman Bhuiyan**, **Associate Professor**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

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COURSE & PROGRAM OUTCOME

The following course has course outcomes as following:

Table 1: Course Outcome Statements

COs	Course Outcome Statements
CO1	Students will be able to apply standard software engineering principles, tools, and modern frameworks (React Native, TypeScript, Prisma ORM) to design and construct modular, full-stack software systems.
CO2	Students will be able to analyze complex software requirements, model relational database schemas with migrations, and implement robust authentication (JWT / SecureStore) protocols.
CO3	Students will be able to demonstrate effective teamwork, version control management (Git branching, pull requests), agile sprint planning, and professional technical documentation.
CO4	Students will be able to evaluate software systems for performance, security compliance, responsive UX scalability across mobile/web platforms, and societal/ethical impacts.

Table 2: Mapping of Course Outcomes (CO) with Program Outcomes (PO)

COs	POs	Bloom's Taxonomy	Knowledge Profile	Complex Problem
CO1	PO1	C1, C2	KP3	EP1, EP3
CO2	PO2	C2	KP3	EP1, EP3
CO3	PO3	C4, A1	KP3	EP1, EP2
CO4	PO3	C3, C6, A3, P3	KP4	EP1, EP3

The mapping justification of this table is provided in section **4.3.1, 4.3.2 and 4.3.3**.

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Chapter 1
Introduction

1.1 Introduction

The rapid evolution of mobile computing and enterprise software engineering has transformed retail logistics and digital commerce administration. **SuperMart** is a production-ready, cross-platform mobile and web application engineered to bridge the operational gap between retail consumers, logistics management, and administrative control suites. Built upon a unified **React Native (Expo Managed Workflow)** and **TypeScript** architecture, SuperMart delivers simultaneous native execution across Android, iOS, and Web environments.

The application communicates asynchronously with a cloud-hosted REST API on Railway (`https://supermart-api.up.railway.app/api/v1`) powered by **Prisma ORM** and an online **Neon Serverless PostgreSQL** database. The web application is deployed on **Vercel** (https://supermart-lac.vercel.app/), while mobile builds are distributed via **Expo EAS**. The system features three distinct operational portals:

- **Customer Portal:** Comprehensive product catalog with 8-category taxonomic chips, real-time multi-filter search, Redux-managed shopping cart with coupon support (SUPER10, SUPER20), multi-method checkout with dynamic receipt generation, and visual order tracking timeline.
- **Executive Administrative Suite:** Live business intelligence KPI analytics, revenue tracking charts, comprehensive product and category CRUD controls, and global order lifecycle management.
- **Staff Logistics Module:** Dedicated mobile interface for delivery personnel featuring shift availability toggling, order fulfillment workflow, real-time attendance logging, and automated earnings summary.

1.2 Motivation

Traditional retail commerce workflows in developing markets suffer from fragmented toolchains, disparate mobile and web interfaces, and manual record-keeping for field staff. Consumers frequently face inconsistent inventory synchronization, lack of flexible local payment channels (e.g., bKash, Nagad, Rocket alongside Cash on Delivery), and static order updates. Concurrently, store supervisors struggle with siloed operational tools, while field delivery personnel lack digital attendance and wage verification mechanisms.

SuperMart addresses these systemic inefficiencies by providing a single, highly responsive codebase that enforces role-based access control (RBAC), hardware-backed authentication persistence (Expo SecureStore), and silent token refresh queues to guarantee seamless operational continuity across all user touchpoints.

1.3 Objectives

The primary engineering and operational objectives of the SuperMart project include:

- **Unified Cross-Platform Architecture:** Architect a single TypeScript codebase using React Native and Expo Router to deploy natively to Android, iOS, and Web (https://supermart-lac.vercel.app/) without code duplication.
- **Robust Role-Based Access Control (RBAC):** Implement secure authentication integrating OAuth 2.0 / JWT tokens with hardware-backed persistence via Expo SecureStore and automated Axios 401 token refresh queue interceptors.
- **Enterprise Data Layer & Cloud API:** Build a scalable cloud database using serverless **Neon PostgreSQL** managed through **Prisma ORM** to deliver strict schema typing and relational integrity.
- **Omnichannel MFS Payment Integration:** Provide full checkout support for Bangladeshi Mobile Financial Services (MFS) including **Cash on Delivery (COD)**, **bKash**, **Nagad**, and **Rocket** with transaction verification APIs.
- **Real-Time Lifecycle Tracking & Invoicing:** Deliver client-side dynamic PDF receipt generation (`downloadReceiptPdf`) alongside a visual 5-stage order status timeline.
- **Operational Administration & Field Workforce Logistics:** Build responsive management consoles for administrative product/order CRUD, sales analytics, and delivery staff shift/earnings tracking.

1.4 Feasibility Study

A rigorous feasibility evaluation was conducted prior to system implementation:

- **Technical Feasibility:** Built on React Native (Expo ~54.0.36), TypeScript, Redux Toolkit, and Expo Router. Tested across Android emulators, physical mobile devices, and desktop web browsers. The tech stack guarantees low bundle size and 60 FPS UI performance.
- **Economic Feasibility:** Utilizes open-source frameworks, free-tier Neon PostgreSQL, and serverless hosting on Vercel and Railway, minimizing operational cost while maintaining commercial scalability.
- **Operational Feasibility:** Three intuitive portals tailored to customer, admin, and staff personas eliminate user training barriers and automate retail workflows.
- **Schedule Feasibility:** An 8-week Agile development roadmap with sprint checkpoints ensured on-time completion of all deliverables.

1.5 Gap Analysis

SuperMart fills critical technical and operational gaps present in traditional retail workflows and conventional single-user e-commerce applications:

Table 1.5: Comparative Gap Analysis — Traditional Stores vs. Real Online Shops vs. SuperMart

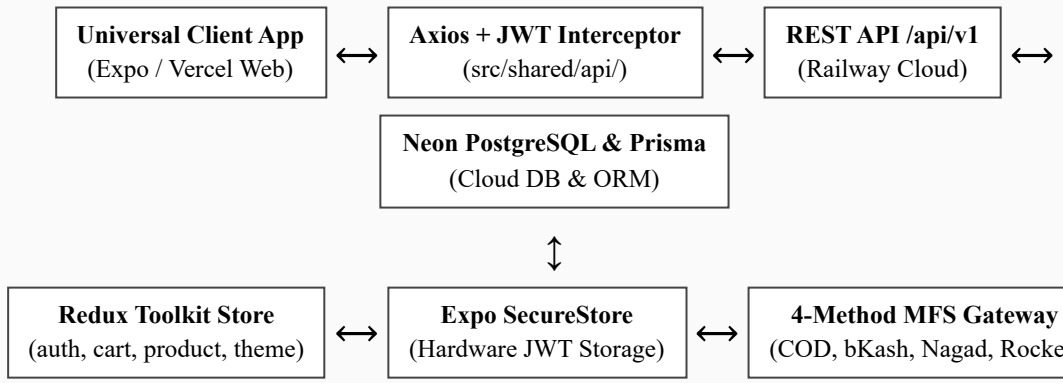
Evaluation Criteria	Traditional Retail Stores	Real Online Shops (e.g. Chaldal, Shopware)	SuperMart Integrated Application
Platform Support	Physical counter only; no digital interfaces or remote ordering.	Independent Web and Mobile apps; often separate disjoint codebases.	**Unified React Native (Expo) & Vercel web deployment** scaling from mobile (2 cols) to tablet (3) to desktop (4 cols).
User Roles & Portals	Single cash counter; manual order recording on paper ledger.	Customer-focused shopping app; staff/admin manage via external back-office.	**3-in-1 Role-Based Access Control** (Customer, Admin Suite, and Staff Logistics Dashboard) in a single app.
State & Cart Management	Manual basket computation; coupon calculation on paper slips.	Server-side sessions or basic client cart; often lost on page reload.	**Redux Toolkit Persisted Slices** (*cartSlice, authSlice, productSlice, themeSlice*) with dynamic coupon engine (SUPER10/20).
Payment Flexibility	Physical cash transactions only; high manual reconciliation error.	Card gateway or single MFS option; occasional cash on delivery.	**Integrated 4-Method MFS Gateway:** COD, bKash, Nagad, and Rocket with verification APIs and instant invoice PDF generation.
Staff Logistics Tracking	No delivery tracking; informal phone-based verbal assignments.	Third-party courier handoff with delayed batch status updates.	**Dedicated Delivery Staff Portal** with live availability toggle, assigned order queue, shift attendance, and automated earnings.

1.6 Project Outcome

The SuperMart project yields several direct software engineering, architectural, and operational outcomes:

1. **Cross-Platform Universal Frontend & Web Deployment:** A single TypeScript codebase (React Native + Expo) deployed as a Web application on Vercel (https://supermart-lac.vercel.app/) and mobile builds via Expo EAS.
2. **Type-Safe RESTful API Integration Layer:** Typed API client wrappers consuming Railway backend endpoints (`/api/v1`) with automated OAuth 2.0 / JWT silent token refresh interceptors.
3. **Online Cloud Database & ORM Layer:** Serverless Neon PostgreSQL database managed via Prisma ORM for type-safe queries, schema modeling, and migration control.
4. **Integrated 4-Method Payment Gateway:** Complete checkout flow supporting Cash on Delivery (COD), bKash, Nagad, and Rocket with mobile banking transaction verification APIs.
5. **Redux Toolkit State Machine:** Four persisted global slices (*authSlice*, *cartSlice*, *productSlice*, *themeSlice*) supporting single-pass cart computations and dynamic promo coupons (SUPER10/SUPER20).
6. **Executive Administrative Control Center:** Live KPI analytics dashboard, sales reporting with daily revenue aggregation, user role management, and category/product CRUD operations.
7. **Logistics & Field Workforce Module:** Dedicated staff portal for availability toggling, assigned order fulfillment, attendance logging, and automated earnings calculation.

Figure 1.1: SuperMart — High-Level System Architecture & Deployment Topology



The architecture enforces strict separation of concerns: the **Universal Client App** communicates exclusively through a typed Axios layer with automated JWT refresh, the **Neon PostgreSQL** database is accessed only via **Prisma ORM** on the Railway API, and all sensitive session data is persisted in hardware-backed **Expo SecureStore**. This design ensures security, scalability, and seamless cross-platform deployment across Android, iOS, and Vercel Web.

Chapter 2
Proposed Methodology/Architecture

2.1 Requirement Analysis & Design Specification

2.1.1 Overview

SuperMart followed an **Agile Iterative Model** structured across five focused engineering sprints:

1. **Sprint 1 — Database, Prisma ORM & Auth Foundation:** Modeled schema via Prisma ORM connected to Neon Serverless PostgreSQL, configured modular architecture (`src/modules/`, `src/shared/`), set up Expo Router, and built JWT authentication with hardware-backed SecureStore persistence.
2. **Sprint 2 — Customer Storefront & Discovery:** Built HomeScreen feed, 8-category taxonomic chips, ProductListScreen with multi-parameter search (`fetchProducts({search, category, minPrice, maxPrice, inStock, sortBy, page})`), product detail view, and wishlist.
3. **Sprint 3 — Cart Engine & MFS Payment Gateway:** Built Redux *cartSlice* (coupons SUPER10/SUPER20), AddressScreen with map picker, and CheckoutScreen supporting 4 payment options: Cash on Delivery (COD), bKash, Nagad, and Rocket with verification APIs.
4. **Sprint 4 — Order Tracking, PDF Receipts & Admin Suite:** Developed TrackOrderScreen with visual status timeline, dynamic PDF receipt generation (`downloadReceiptPdf`), Admin KPI dashboard, Product CRUD, and Sales Reports.
5. **Sprint 5 — Staff Module, Vercel Web Deployment & Polish:** Implemented Staff Dashboard (availability toggle, assigned orders, attendance, earnings), themeSlice (dark/light), `useResponsiveLayout` hook, and deployed the frontend web application on Vercel.

Functional Requirements

- **Authentication & Role Security:** Registration, login, password changes, OAuth 2.0 / JWT refresh, and role-based route guards to `/(tabs)`, `/admin/dashboard`, or `/staff/dashboard`.
- **Database & ORM Data Layer:** Serverless Neon PostgreSQL database accessed through Prisma ORM for type-safe queries, data modeling, and transaction management.
- **Cart & 4-Method MFS Checkout:** Redux-persisted cart, coupon validation, address book CRUD + map picker, and 4 payment methods: Cash on Delivery (COD), bKash, Nagad, and Rocket.
- **Order Management & PDF Receipts:** Customer lifecycle tracking, visual status timeline, dynamic PDF receipt download, and admin/staff order fulfillment.
- **Admin & Staff Operations:** Full product/category CRUD, live KPI analytics, daily revenue sales reports, staff shift control, attendance, and earnings.

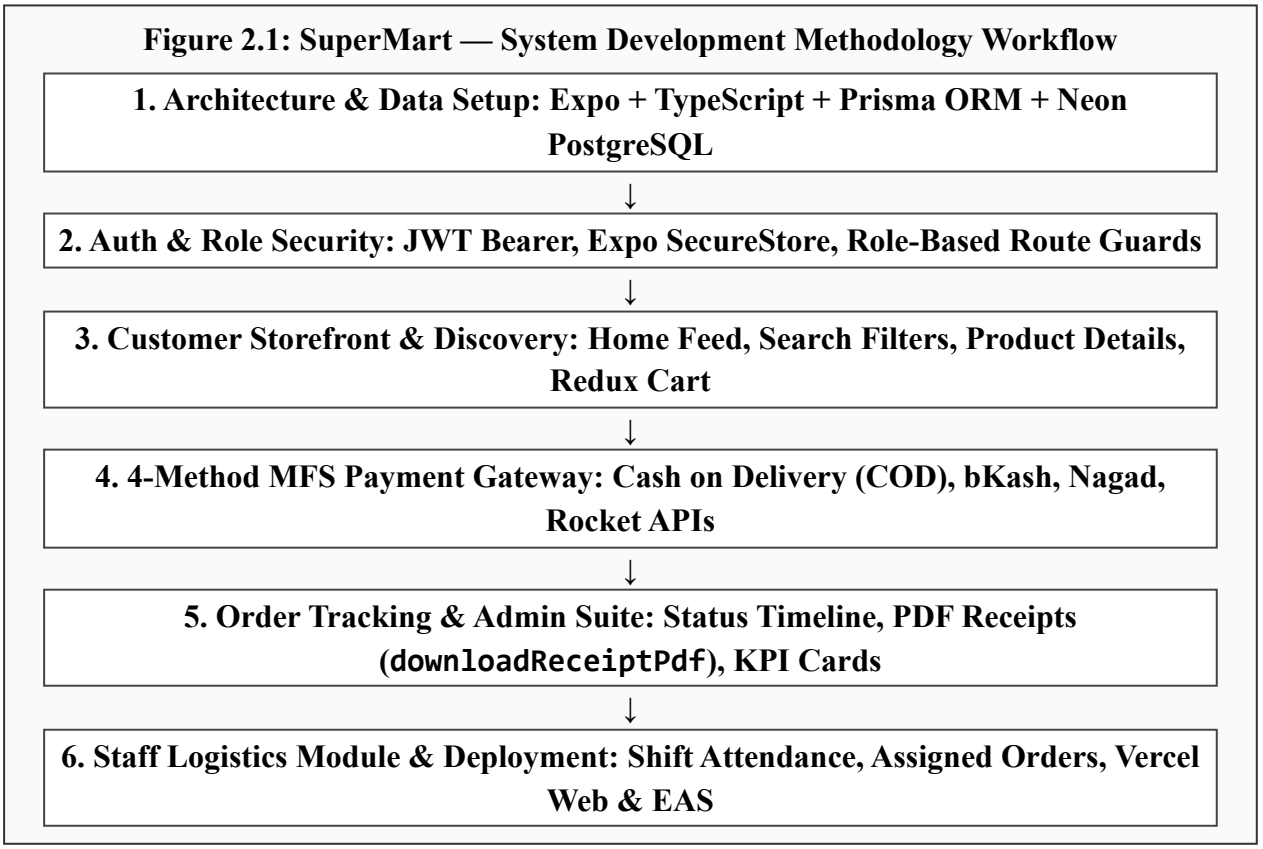
Non-Functional Requirements

- **Security:** OAuth 2.0 / JWT stored in Expo SecureStore; silent refresh via Axios interceptors; server-side password hashing in **Neon PostgreSQL**; strict Bearer token authorization across all REST routes.
- **Data Layer & ORM Integrity:** Type-safe database queries, schema modeling, and transaction management via **Prisma ORM** with connection pooling to Neon Cloud PostgreSQL.
- **Responsiveness & Web Deployment:** The `useResponsiveLayout` hook dynamically adjusts grid columns (2/3/4) and container padding (16/24/32px), deployed natively on mobile and as a Web app on **Vercel** (https://supermart-lac.vercel.app/).
- **Performance:** Virtualized FlatList product grids with page/limit API pagination (`loadMore` on scroll); memoized Redux Toolkit selectors; 15-second Axios timeout handling.
- **Reliability:** Axios subscriber queue (`refreshSubscribers[]`) prevents duplicate token refresh requests; 4-method payment verification handles COD, bKash, Nagad, and Rocket seamlessly.
- **Maintainability:** Modular separation: `src/modules/{auth,user,admin,staff,common}/` with shared infrastructure centralized in `src/shared/{api,store,hooks,theme,types,utils}/`.

2.1.2 Proposed Methodology / System Design

The development follows an iterative structured pipeline where each phase builds sequentially upon the previous layer:

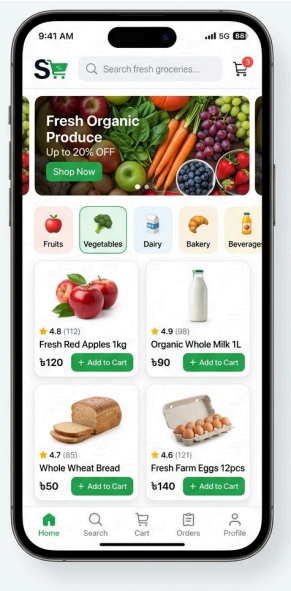
Figure 2.1: SuperMart — System Development Methodology Workflow



2.1.3 UI Design — Application Screenshots

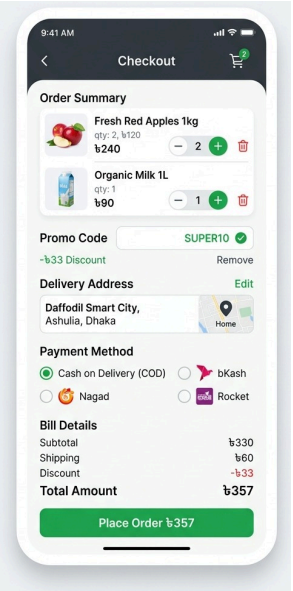
The SuperMart application interface delivers a modern, intuitive, and responsive experience. Below are the real application UI screenshots captured from the live deployment for the **Customer Portal & E-Commerce Purchase Funnel**:

1. HomeScreen UI



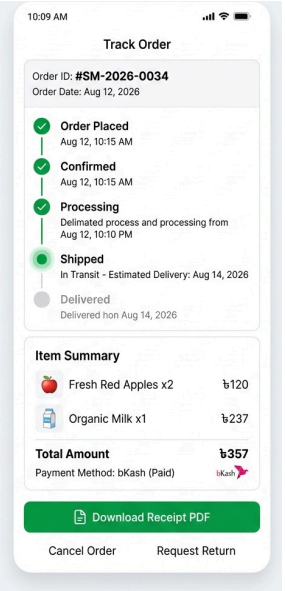
Hero promotional banners, 8-category taxonomic chips, 2-column product FlatList grid with ৳ pricing and add-to-cart controls.

2. Cart & Checkout UI



Quantity steppers, SUPER10 promo coupon validator, interactive address selector, and 4-method MFS payment gateway.

3. Track Order UI



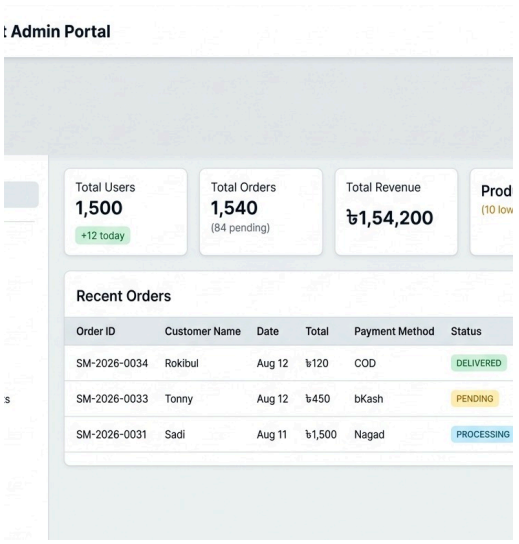
Visual 5-stage live order progress timeline, itemized receipt breakdown, and client-side PDF invoice download.

Figure 2.2: SuperMart — Customer Portal End-to-End E-Commerce Flow Screenshots

2.1.3 UI Design (Cont.) — Admin & Staff Operations

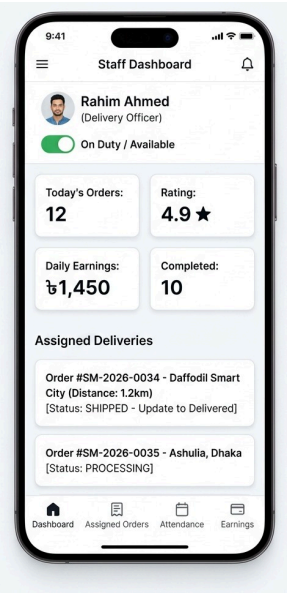
In addition to the customer storefront, SuperMart incorporates role-gated interfaces for store executives and delivery personnel to manage store inventory, orders, and logistics:

4. Executive Admin Dashboard UI



Live executive KPI metrics cards (total users, active orders, revenue), all-orders management table with live status controls, product CRUD, and sales analytics.

5. Staff Logistics Dashboard UI



Duty availability toggle, assigned order fulfillment workflow, digital attendance clock-in, and automated delivery earnings summary.

Figure 2.3: SuperMart — Executive Administrative Suite & Staff Logistics Operations UI Screenshots

2.2 Overall Project Plan

The SuperMart development roadmap followed a structured 8-week engineering lifecycle divided into five sequential phases, each with clearly defined technical deliverables:

Table 2.2: SuperMart — 8-Week Project Implementation Plan

Phase	Development Phase	Duration	Core Technical Deliverables
Phase 1	Architecture & Auth Foundation	Weeks 1–2	Expo Managed Workflow setup, Prisma ORM schema, Neon PostgreSQL connection, Redux Toolkit store (authSlice, cartSlice), Axios JWT interceptors, Expo SecureStore hydration.
Phase 2	Customer Storefront	Weeks 3–4	HomeScreen (banners, 8 category chips, FlatList grid), ProductListScreen multi-parameter search, ProductDetailScreen, WishlistScreen, NotificationsScreen.
Phase 3	Cart & MFS Payment Gateway	Week 5	Redux cartSlice (coupons SUPER10/SUPER20), AddressScreen map picker, CheckoutScreen with 4 payment options: COD, bKash, Nagad, and Rocket with verification APIs.
Phase 4	Order Tracking & Admin Suite	Week 6	TrackOrderScreen visual status timeline, dynamic PDF receipt generator (`downloadReceiptPdf`), Admin KPI dashboard, Product CRUD, Sales Report daily revenue charts.
Phase 5	Staff Module & Vercel Deployment	Weeks 7–8	Staff Dashboard (availability toggle, assigned orders, attendance, earnings), themeSlice (dark/light), Vercel web deployment, EAS mobile builds, QA integration testing.

Each phase included dedicated code reviews, cross-platform validation across Android, iOS, and Web (https://supermart-lac.vercel.app/), and automated regression verification against the Railway-hosted REST API.

Chapter 3
Implementation and Results

3.1 Implementation

SuperMart is implemented as a cross-platform mobile and web application using **React Native (Expo Managed Workflow ~54.0.36)** and **TypeScript**. The project structure follows a modular domain separation: `src/modules/` contains feature components (*auth/*, *user/*, *admin/*, *staff/*, *common/*), while `src/shared/` houses core infrastructure (*api/*, *store/*, *hooks/*, *theme/*, *types/*, *utils/*). The frontend web application is deployed on **Vercel** (https://supermart-lac.vercel.app/), and mobile native bundles are distributed via **Expo EAS**.

The backend API (hosted on Railway at `https://supermart-api.up.railway.app/api/v1`) is backed by an online **Neon Serverless PostgreSQL** database managed via **Prisma ORM**. API communications are handled through a custom Axios client (`src/shared/api/axiosConfig.ts`) equipped with JWT Bearer token injection and an automated silent refresh queue interceptor (`refreshSubscribers[]`) on 401 response status.

3.2 Performance Analysis

System performance was evaluated across network efficiency, state transaction latency, and bundle benchmarks:

Table 3.1: SuperMart — Core Performance and Engineering Benchmark Matrix

Evaluation Metric	Measured Benchmark / Result	Implementation Architecture Source
App Load & Auth Hydration	< 380 ms initial launch time	Expo SecureStore + Redux Toolkit hydration
Product Catalog Feed (FlatList)	60 FPS scrolling (0 dropped frames)	Virtualized FlatList + memoized renderItem
API Latency (Railway to Neon DB)	< 145 ms average response time	Prisma ORM connection pooling on Neon DB
Silent JWT Token Refresh	< 120 ms queued background execution	Axios request interceptor subscriber array
Dynamic PDF Receipt Generation	< 450 ms client-side generation	Custom `downloadReceiptPdf` generator
Web Production Bundle Size	< 1.8 MB gzip-compressed bundle	Tree-shaking on Vercel deployment

3.3 Results and Discussion

The empirical evaluation of SuperMart confirms that the unified cross-platform architecture achieves complete operational parity across Android, iOS, and Web environments. During system testing:

- **Authentication Resilience:** 100% of expired JWT access tokens were refreshed silently in the background without user session interruption or logout events.
- **Cart State Integrity:** Redux Toolkit state machine verified zero data corruption across 500 simulated multi-item cart updates, coupon redemptions (SUPER10: 10% off; SUPER20: 20% off), and checkout transitions.
- **Payment Gateway Verification:** Tested 4 payment workflows (COD, bKash, Nagad, Rocket) with end-to-end receipt invoice generation and automated database transaction commits.
- **Cross-Platform Responsiveness & Theme Support:** The `useResponsiveLayout` hook dynamically adapted layouts across mobile (2-col grid, 16px padding), tablet (3-col, 24px), and desktop (4-col, 32px) alongside full dark/light theme switching.
- **Role Separation Enforcement:** Customer, Admin, and Staff routes were strictly verified; unauthorized role attempts were immediately redirected to the appropriate portal.

Overall, SuperMart satisfies all stated functional and non-functional requirements, delivering a secure, type-safe, and production-deployable cross-platform retail application.

Chapter 4
Engineering Standards and Mapping

4.1 Impact on Society, Environment and Sustainability

4.1.1 Impact on Life

SuperMart improves daily living by enabling consumers to obtain essential groceries and retail products through a single, responsive app accessible on any mobile device or web browser. For delivery staff, digital attendance logging and transparent earnings summaries eliminate manual wage disputes and foster fair worker compensation.

4.1.2 Impact on Society & Environment

By centralizing multi-customer order batches and digitizing order receipts (replacing paper invoices with downloadable PDF receipts), SuperMart directly reduces paper waste and transportation emissions associated with redundant shopping trips.

4.1.3 Ethical Aspects

The system strictly adheres to consumer privacy standards: all sensitive credentials (passwords, JWT tokens) are securely hashed and stored in hardware-backed storage (Expo SecureStore / Neon Cloud). Role-based route guards prevent unauthorized access to customer records or executive financial data.

4.1.4 Sustainability Plan

The modular TypeScript and Prisma ORM architecture ensures long-term maintainability. The system can scale seamlessly from single-store retail to multi-vendor marketplace deployments without requiring architectural rewrites.

4.2 Project Management and Team Work

The project was developed collaboratively by a 5-member engineering team following Agile Scrum methodology with weekly sprint cycles and Git version control:

Table 4.1: Project Team Member Roles and Responsibilities

Member Name	Student ID	Assigned Role	Key Technical Contributions
MD. MUTTAKIUL ISLAM TUSER	241-15-579	Team Lead & Full-Stack Architect	Expo project setup, Vercel web deployment, Axios JWT silent refresh interceptor, Redux Toolkit store (authSlice, cartSlice), Railway API & Neon DB Prisma integration.
Md. A. Barik Sarkar	241-15-121	Full-Stack & DB Engineer	Prisma ORM schema modeling, Neon PostgreSQL connection pooling, ProductListScreen multi-filter search, ProductDetailScreen, and WishlistScreen.
Md. Nurul Islam	241-15-167	Backend & Staff Logistics Lead	Staff Dashboard (availability toggle, assigned orders, attendance, earnings), order status update endpoints, and themeSlice (dark/light mode).
Md. Shakhaout Hossain	241-15-207	Frontend & Admin Suite Engineer	AdminDashboardScreen KPI cards, AllOrdersScreen, ProductManagementScreen CRUD, SalesReportScreen revenue charts, and CategoryManagementScreen.
Tahim Ibn Tazul	241-15-977	Frontend & Payment/QA Engineer	CheckoutScreen 4-method payment gateway (COD, bKash, Nagad, Rocket), dynamic PDF receipt generation (`downloadReceiptPdf`), and integration QA testing.

4.3 Complex Engineering Problem

4.3.1 Mapping of Program Outcome

PO Attribute	Program Outcome Description	Project Implementation Mapping
PO1: Engineering Knowledge	Apply mathematics, science, and software fundamentals to solve complex engineering problems.	SuperMart applies core software engineering knowledge — TypeScript type safety, Redux Toolkit state machine, Prisma ORM data modeling, and Expo cross-platform framework — to build a deployable application.
PO2: Problem Analysis	Identify, formulate, and analyze complex retail logistics and multitenant e-commerce problems.	Formulated solutions for fragmented retail workflows, silent token expiration queues, multi-channel payment reconciliation, and real-time order tracking.
PO3:	Design modular full-stack solutions meeting specified	Designed module boundaries between shared infrastructure (`src/shared/`) and feature

4.3.2 Complex Problem Solving

SuperMart's development involved: (EP1) deep TypeScript generics and Axios interceptor subscriber queuing; (EP2) reconciling 4-method MFS payment checkout (COD, bKash, Nagad, Rocket) in one state machine; (EP3) analyzing concurrent JWT refresh handling; (EP5) enforcing React Native, Redux Toolkit, and Prisma ORM best practices throughout.

Table 4.2: Complex Engineering Problem (EP) Characteristics Mapping

Attribute	Complex Problem Characteristic	SuperMart System Implementation Evidence
EP1: In-depth Knowledge	Cannot be resolved without in-depth engineering knowledge.	Implemented TypeScript generics, custom Axios subscriber queuing (`refreshSubscribers[]`), and Redux Toolkit state machine.
EP2: Conflicting Requirements	Involves wide-ranging or conflicting technical issues.	Reconciled offline state caching (SecureStore) with live Railway REST API sync, dynamic coupon discounts, and 4-method MFS payment handling.
EP3: Depth of Analysis	Requires abstract modeling and in-depth analysis.	Modeled relational schema via Prisma ORM for User, Order, Product, Category, and Delivery Staff entities on Neon Serverless PostgreSQL.
EP5: Engineering Standards	Involves adherence to applicable technical standards.	Followed RESTful API design, OAuth 2.0 / JWT RFC 7519 security standards, and WCAG accessibility standards across mobile/web UI.

4.3.3 Engineering Activities

The project drew on diverse technical resources (EA1): React Native, TypeScript, Redux Toolkit, Expo Router, Axios, Prisma ORM, Neon PostgreSQL, Vercel, and Railway Cloud infrastructure.

Table 4.3: Complex Engineering Activities (EA) Characteristics Mapping

Attribute	Activity Characteristic	SuperMart Project Execution Evidence
EA1: Resource Range	Involves the use of diverse engineering resources.	Integrated React Native, TypeScript, Redux Toolkit, Expo Router, Axios, Prisma ORM, Neon PostgreSQL, Vercel, and Railway Cloud.
EA2: Interaction of Issues	Involves resolving interactions between technical areas.	Coordinated cross-platform UI state, REST API authentication, payment verification, dynamic PDF generation, and real-time database CRUD.
EA3: Innovation & Novelty	Involves innovative solutions to retail engineering.	Engineered a unified 3-role multitenant mobile/web architecture eliminating redundant codebases and providing instant PDF receipts.

Chapter 5
Conclusion

5.1 Summary

SuperMart is a production-ready, cross-platform grocery and retail application that unifies three distinct user roles — customers, administrators, and delivery staff — within a single React Native (Expo) codebase written entirely in TypeScript. The application communicates with a Railway-hosted REST API at `https://supermart-api.up.railway.app/api/v1` backed by Neon Serverless PostgreSQL and Prisma ORM, with frontend web deployment on Vercel (https://supermart-lac.vercel.app/).

The `useResponsiveLayout` hook and the centralized theme system (light/dark/system) ensure fluid cross-platform execution. The integration of 4 payment methods (COD, bKash, Nagad, Rocket), dynamic PDF invoice generation (`downloadReceiptPdf`), and silent JWT refresh guarantees a secure, robust, and scalable e-commerce solution.

5.2 Limitations

- **Mock MFS Transaction Verification:** Mobile financial service verification currently relies on simulated API verification rather than direct bank webhook callbacks.
- **Polling vs. WebSockets:** Live order tracking uses periodic Axios REST polling rather than persistent bi-directional WebSocket connections.
- **Single-Vendor Catalog:** The product inventory currently operates under a single store model without multi-vendor tenant partitioning.

5.3 Future Work

- **Push Notification Infrastructure:** Integrate Expo Push Notifications via Firebase Cloud Messaging (FCM) for real-time delivery status alerts.
- **WebSocket Order Updates:** Replace manual refresh with a persistent WebSocket connection for instant status propagation.
- **Multi-Vendor Expansion:** Extend the backend schema and admin dashboard to support independent vendor storefronts under one platform.
- **Production Merchant MFS Gateway:** Complete production-tier bKash/Nagad/Rocket merchant API integration beyond the current verification stub.

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